Grid Inverter Software Specification - Inresol

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(Commercial In Confidence)

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Revision Change Notes:

Grid inverter Software Specification - Inresol

1 Scope

This proposal covers specifications for the development of software for a 10kW 3-phase Grid Inverter for Inresol.

2 Summary and Recommendations

This document covers description and specification for the software and operation of the grid inverter for Inresol from review and discussion within the SDC team in Egham.

The project is at an early stage of development and it is therefore anticipated that changes will be made to these requirements as further review is undertaken, progress is made and experience and learning is achieved in key areas.

Operation under Phase Locked Loop with the grid is identified as crucial. From discussion with Microchip it is understood that much of this software capability will need to be created from scratch. However, the techniques identified in the sections covering PLL are commonly used with motor drive techniques. Clarke and Park transforms are used for control of 3 phase motors, for example. As such we anticipate that there is good knowledge in Microchip to support development forward and we have confidence that the selected dspic part will have sufficient capability.

Synchronous-Frame (SF-PLL) is identified as the preferred PLL method, due to its relative simplicity and good performance. This is reportedly widely used across the inverter industry. However, we should keep an open mind towards implementing other PLL schemes, if benefits are clear and proven.

The structure of the state machine described has a default idle state, which is returned to if changes between local load and grid connection are necessary or instructed. The structure of the state machine is done this way to ensure safe operation.

Recommendations:-

1. This specification is reviewed by all concerned parties, as a next step.
2. Software development begins on both the overall state machine operation and PLL development in parallel.

3 System Operation and Overview

The 3-phase grid inverter is part of a power generation system being developed by Inresol which uses a stirling engine as the primary power source. This system may supply power directly to the grid (single or 3-phase) or to a local load which is off-grid.

The maximum power capability of the system is 10kW.

The grid inverter for this power generation system connects directly with the grid or the local load and converts an input voltage DC source to single or 3-phase AC output.

Figure 1 below shows a block diagram for the full system.

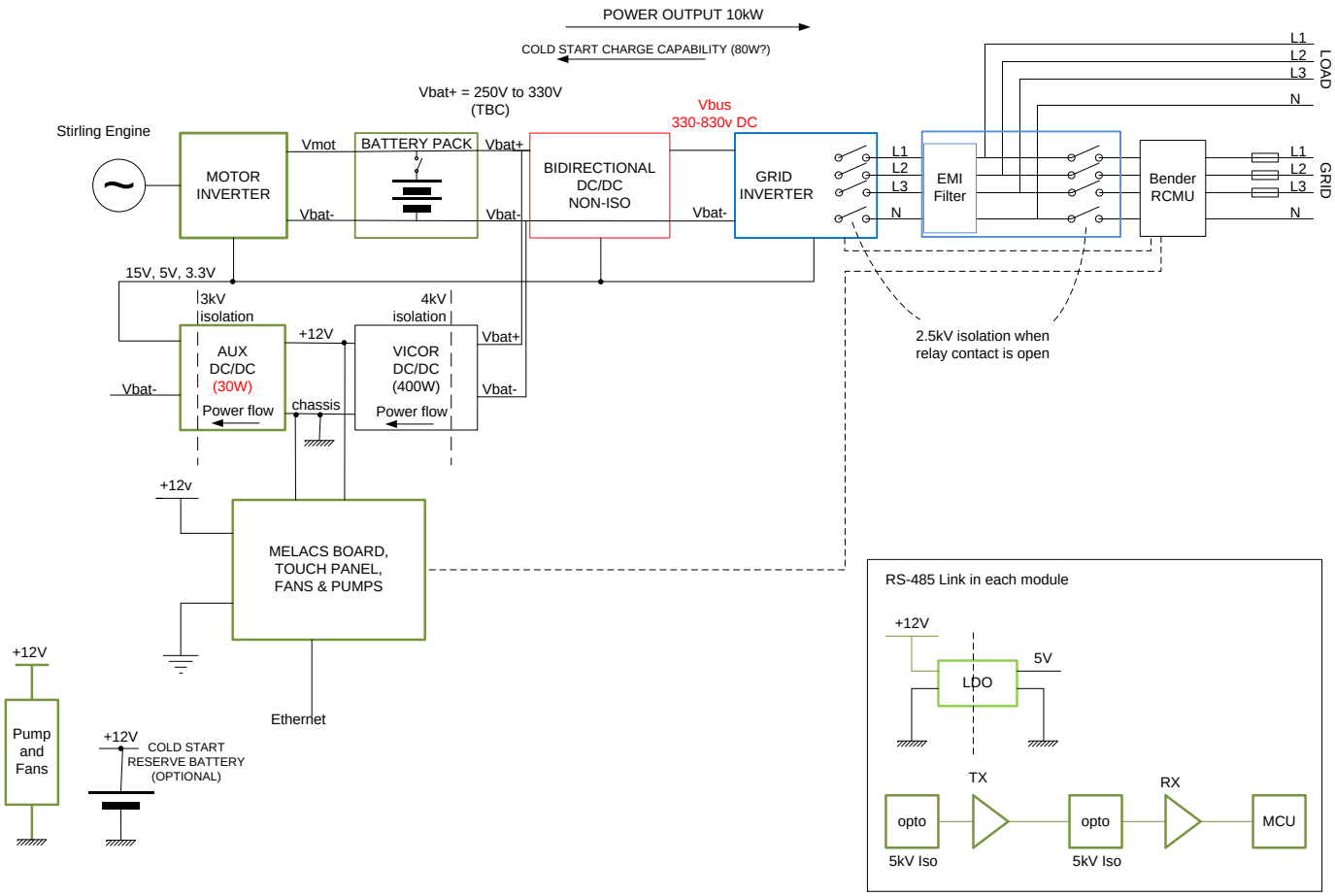


Figure 1: Stirling Engine power Generation System.

The full system consists of the following functional sub-systems:-

- **3-phase generator** attached to Stirling Engine.
- **Motor Inverter:** which converters the 3-phase output from the stirling engine generator to a regulated DC voltage and charges the main battery pack.
- **Battery pack:** This is a 1kWh LiFePO4 battery pack. The output voltage can vary from 250V to 330V.
- **Bi-directional non-isolated DC-DC stage:** This stage boosts the output from the battery pack to a useable bus voltage level (ranging from 330V to 830V, depending on the output AC supply needed).
    - During start-up this stage will also operate as a buck converter to charge the battery pack from an AC source connected via the grid inverter.
- **Auxilliary Power Supplies:** To power pumps and fans and low voltage circuits.
- **Central Control Board:** Referred to as the 'Melacs' board by inresol. This board allows user interface and display, communication and overall control of the respective sub-systems.
- **Grid Inverter and EMI filter:** Converters DC voltage input from the DC-DC converter to AC output (local or grid).

To date, Inresol have requested development support for the grid inverter and the DC-DC converter. Our proposal for development of a grid inverter was agreed last year and first pass hardware design is nearing completion. Schematics are available for review.

Our proposal for development of the DC-DC converter remains under review.

3.1 Communications

Communication across the system to and from the Melacs board is via isolated RS-485[1]. Direct communication between the DC-DC stage and the grid inverter is accommodated via UART connection.

3.2 System Operation

The stirling engine operates directly from heat exchange and as such can use any combustible material as a fuel source, or a direct source of heat (e.g. from industrial process exhausts). The generator attached to the stirling engine creates a varying frequency and varying voltage AC output.

The motor inverter receives the AC input from the stirling engine generator and outputs a regulated DC voltage allowing charge of the main battery pack.

The battery pack is 1kWh in size and has a voltage varying between 250V and 330V, depending on its state of charge. The battery pack is used as a reserve power source should the stirling engine need resetting, refuelling or servicing for a short period. It also provides sufficient power to kick-start the stirling engine (since there is inertia). Inresol have reported that a reserve energy level between 0.3kWh and 0.5kWh is required to start the stirling engine.

In addition to the above, the battery voltage feeds the auxiliary power supplies in the system to deliver the low voltage power supply (+12V to chassis (earth)). This low-voltage power supply powers the pumps and fans for the system as a whole, as well as the Melacs board and user interface panel.

[1] A preferred protocol has not yet been defined.

Power for the control electronics for the motor inverter, DC-DC converter and grid inverter is also derived from the low-voltage +12V power supply via an isolated DC-DC converter.

3.3 Cold Start and Main Battery Charge

If the main battery pack is not charged to a level sufficient to kick-start the stirling engine then it needs to be charged from an external power connection. This will either be the AC grid or, in the case of no grid being present, an external AC inverter fed from a DC source (e.g. 12V car battery).

To charge the battery pack the grid inverter acts as a passive rectifier, using the body diodes/ co-packaged diodes within the IGBT devices. The controlled switching is undertaken by the bi-directional DC-DC converter. In this mode of operation the grid inverter controls its relays, monitors voltages and currents and reports with the Melacs board and DC-DC converter.

A reserve +12V battery is available within the system to provide power to the control electronics, should the main battery pack state of charge be so low that it could not power anything safely, causing itself permanent damage (i.e. the voltage level is below 250V).

3.4 Operation Under Load

When supplying power to an external load, 4 modes of operation are possible:-

1. Single phase supply off-grid
2. 3-phase supply off-grid
3. Single phase supply to grid
4. 3-phase supply to grid

The maximum load which can be supplied single phase is 3.33kW and the maximum load which can be supplied 3-phase is 10kW.

3.4.1 Supplying Power to the Grid

It is assumed that the grid inverter will receive information from the central control board to determine the level of power that should be supplied to the grid[2]. This level will depend on the power available and being generated by the stirling engine. As such operation is dependent on a higher level control loop utilising the RS-485 communications within the overall system.

Depending on the location of the equipment, there may be additional requirements which the grid inverter needs to comply with. These will include the maximum allowable power that can be supplied to the grid (this may differ to the rated power level of the equipment), and the mode of operation should a fault occur on the grid[3].

At the time of writing, it is assumed that the level of power that should be supplied to the grid will be updated by the central control board on a regular basis. At this time a 1 second update for this information appears reasonable.

It is important for the reader to note that, under grid connection, voltage from the inverter is synchronised and slightly higher with the grid, and that it is alternating current which is controlled to deliver the required power to the grid. The local load, since it is connected to the grid, is also part of the grid.

[2] At the time of writing this method of operation is assumed.
[3] These requirements have yet to be defined.

Default grid inverter behaviour in the event of a fault on the grid is assumed i.e. so-called 'passive' anti-islanding. Further details on this default operation are provided in section 5.6.

The grid inverter will also need to be aware of the level of the main battery pack voltage. This information can be made available either via the Melacs board over RS-485, or directly from the DC-DC converter. At the time of writing it is assumed that this information will be made available directly from the DC-DC converter. Should the battery voltage drop to 250V then supply of power to the grid (or local load) should be stopped, in order to safeguard the main battery pack from permanent damage.

3.4.2 Supplying Power to a Local Load i.e. Off-Grid

When delivering power to the local load, power demand is determined by the local load.

The grid inverter will be required to report load demand over RS-485 on a regular basis, (a 1 second interval is assumed) so that the outer system control loops can monitor and adjust power generation from the stirling engine.

Under local load supply, the grid inverter will control the output AC voltage and frequency within required limits, whilst current demand is determined directly by the local load. The control loop to achieve this operation differs from the grid connection control loop.

3.4.3 Changing Operation Should the Grid Appear

If the grid is detected, whilst supplying the local load, a safe transition of operation is required as a priority.

If the grid appears present and stable for a defined period (nominally 5 seconds), then this should be communicated to the central control board.

The Melacs board will respond to confirm grid connection (only the Melacs board can control and give permission for grid connection).

From this point there is a choice of operation:-

1. Cease supply to the load. Revert to a safe start state. Start-up inverter ready for grid connection. Apply grid connection.
2. Continue supply to local load. Whilst monitoring the grid, slowly adjust phase and voltage so that a controlled connection to the grid is possible. Switch control loop from voltage control to current control. Connect with grid.

Option 1 above is currently recommended and defined by the state machine (section 4).

3.4.4 Changing Operation Should the Grid Disappear or a Grid Fault is Detected (Anti-Islanding)

At the time of writing it is not certain what behaviour is required should a fault occur on the grid and/ or grid connection is lost. Future Electronics is aware only that regulations are being reviewed (section 5.6).

There is, however, a default mode of operation and this is recommended for implementation at the time of writing. If connection to the grid is lost, then the grid inverter is likely to observe an increase in load demand and/or a decrease in grid voltage. Should these occur beyond defined limits, then a grid fault is detected.

From this point there is a choice of operation:-

1. Cease supply to the grid. Revert to a safe start state. Start-up inverter ready for local load connection. Apply load connection.

2. Switch control loop from current control to voltage control. Connect to local load.

Option 1 is currently recommended and defined by the state machine in section 4.

4 Grid inverter Hardware

A simplified block diagram taken from the grid inverter schematics is shown below in figure 2. This highlights connections and signals to and from the dsPIC microcontroller and the basic functionality of the grid inverter itself.

The input to the grid inverter (UDC_P – UDC_N) is connected to the bus voltage level provided by the DC-DC converter. This will range from 330V to 830V, depending on the AC output requirements.

4.1 UDC_P and UDC_N Required Levels

Table 1 below list the nominal input voltages required for different AC output requirements (Vbus = 3.4 x AC voltage)

AC Output (50Hz or 60Hz) (V)	Nominal Vbus Grid inverter input (UDC_P – UDC_N) (V)
100	340
115	390
120	408
127	432
220	748
230	782
240	816

Table 1: nominal input voltage levels to grid inverter.

4.2 Inverter Switching Operation

The inverter switching is completed by 3 IGBT half-bridges. The bridges switch at a frequency of 16kHz (nominal). When all 3 half-bridges are operating the output is 3-phase. When one half-bridge is switching the output is single-phase. The LCL filter filters the switched waveform to the required 50Hz or 60Hz. Further filtering is applied by the EMI filter to meet harmonic requirements for grid connection.

To output power to the local load relays K103, K200, K300 and K400 are closed.

To output power to the grid the relays above and relays K104, K100, K101 and K102 are closed.

4.3 ADC Measurements

Voltage is measured at the following points:-

- Input Vbus voltage (UDC).

- The output of the inverter before relays K200, K300, K400 (U_INV).
- The output of the inverter after relays K200, K300, K400 (U_MID).
- Grid voltage (U_GRID).

When operating via voltage control (i.e. supplying local load) U_INV and U_MID are the measurements used for control.

Current is measured at the following points:-

- The output of the inverter (I_INV).
- The output of the inverter after relays K200, K300, K400 (I_MID).
- Grid voltage (I_GRID)
-

When operating via current control (i.e. when connected to the grid) I_INV are the measurements used for control.

Temperature inputs are provided from the following points in the system:-

- IGBT heatsink
- LCL filter
- Ambient

These are the inputs to the ADCs on the dsPIC.

4.4 Voltage and Current Scaling

The following sections detail the scaling in the hardware for Voltage and current measurements in the system.

4.4.1 U_INV and U_MID

Max input voltage defined as the peak value of 240Vac +10% (upper grid tolerance) +10% safety margin.

$240Vrms * 1.1 * \sqrt{2} * 1.1 = 410V$

Measured value (w.r.t. M)	scaled down to dsPIC ADC Input
410V	3.147V
0V	1.65V
-410V	0.177V

$$Gain = \frac{410V - (-410V)}{3.147V - 0.177V} = 276.1$$

4.4.2 U_GRID

Max input voltage defined as the peak value of 240Vac +10% (upper grid tolerance) +10% safety margin. We need to be able to detect a mis-wiring leading to connecting a phase to neutral. So $\sqrt{3}$ added.

$240Vrms * 1.1 * \sqrt{2} * 1.1 * \sqrt{3} = 711V$

Measured value (w.r.t. N_Grid)	scaled down to dsPIC
711V	3.214V
0V	1.65V
-711V	0.110V

$$Gain = \frac{711V - (-711V)}{3.214V - 0.110V} = 458.1$$

4.4.3 U_UDC_P and U_UDC_N

Max input voltage defined as the max dc link voltage allowed per capacitor +20%.

$450Vdc * 1.2 = 540V$

Measured value	scaled down to dsPIC
540V	2.831V
0V	0V

$$Gain = \frac{540V - 0V}{2.831V - 0V} = 190.7$$

4.4.4 I_INV and I_GRID

Output from ACS716 plus opamp gain.
1.65V = 0A

$$18.5\frac{mV}{A} * \frac{6k8 + 10k}{10k} = 31.08\frac{mV}{A}$$

4.5 Remaining Signals

4 pairs of PWM output are used. 3 are used to drive the IGBT half-bridges respectively. A 4th PWM pair is used to drive the power supply for the IGBT drive circuitry.

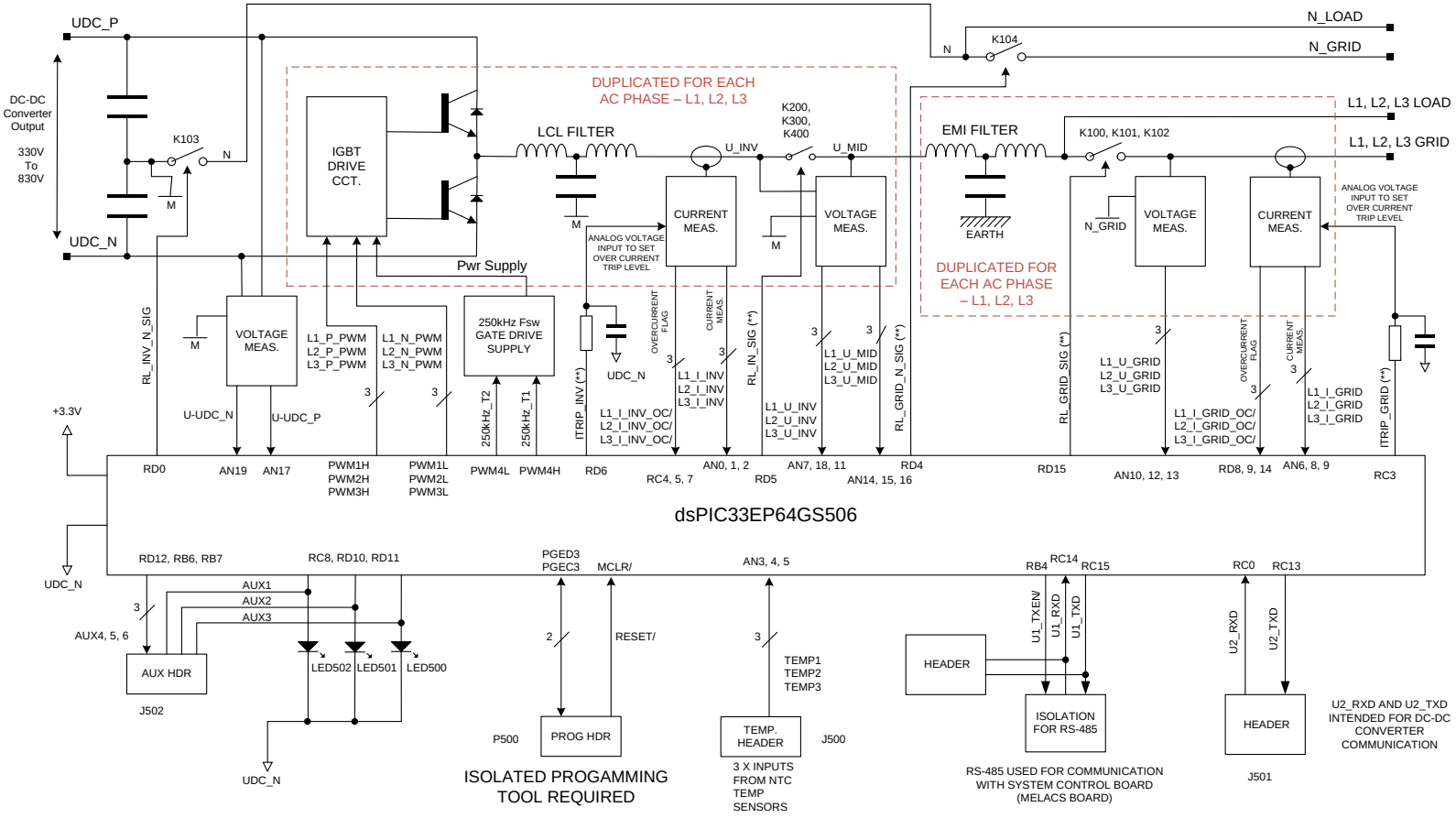
Several GPIOs are used to control the relays and flag errors.

2 UARTs are used, one for local connection with the DC-DC converter and one for RS-485 communication across the system.

6 GPIO pins are spare (3 are connected to LEDs for debug purposes).

Programming interface is provided by the PGED and PGEC inputs.

Table 2 (below) provides a full list of signals with the dsPIC.



NOTE: (**) IDENTIFIES SIGNALS WHICH ARE COMMON ACROSS ALL AC PHASE CIRCUITS

NOTE: HEADERS MUST BE GALVANICALLY ISOLATED FROM SUPPORTING TEST/ PROGRAMMING EQUIPMENT

Figure 2: Grid Inverter Hardware and connections

Signal	MCU	Pin	Description	Type	Purpose	Timing	Group
RL_INV_N_SIG	RD0	47	Opens/Closes Relay K103 Hi = Relay Closed	GPIO (Output Only)	Connects Neutral to M. M is the midpoint potential between Vbus and Vbat- and also midpoint potential for inverter AC phases.	Hi during operation, once AC phase voltages from inverter have reached required level and synchronisation with any external grid AC phases. Pulled Lo on start-up or safety trip/ shutdown. Default/ Reset = Lo	Relay
RL_IN_SIG	RD5	53	Opens/ Closes Relays K200, K300, K400 Hi = Relays Closed	GPIO (Output Only)	Connects AC phases from inverter (U_INV) through to EMI filter and local AC loads (U_MID).	Hi during operation, once AC phase voltages from inverter have reached required level and synchronisation with any external grid AC phases. Pulled Lo on start-up or safety trip/ shutdown. Default/ Reset = Lo	Relay
RL_GRID_N_SIG	RD4	58	Opens/Closes Relay K104 Hi = Relay Closed	GPIO (Output Only)	Connects Grid Neutral to load neutral/ inverter neutral	Hi when power output to the grid is required, once inverter is operating and running correctly. Pulled Lo when only local load supply needed or for safety trip/ shutdown. Default/ Reset = Lo	Relay
RL_GRID_SIG	RD15	59	Opens/ Closes Relays K100, K101, K102 Hi = Relays Closed	GPIO (Output Only)	Connects Grid AC phases to load/ inverter AC phases	Hi when power output to the grid is required, once inverter is operating and running correctly. Pulled Lo when only local load supply needed or for safety trip/ shutdown. Default/ Reset = Lo	Relay
U-UDC_N	AN19	46	Analog input from voltage measurement for input bus voltage (DC)	ADC Input	Samples the measurement output for voltage UDC_N w.r.t voltage M.	All the time at a periodic rate (nominally once every second). (System operation will flag to the DC-DC Converter if UDC_P - UDC_N becomes too large or too small).	ADC IN Voltage
U-UDC_P	AN17	52	Analog input from voltage measurement for input bus voltage (DC)	ADC input	Samples the measurement output for voltage UDC_P w.r.t voltage M.	All the time at a periodic rate (nominally once every second). (System operation will flag to the DC-DC Converter if UDC_P - UDC_N becomes too large or too small).	ADC IN Voltage
L1_U_INV	AN7	29	Analog input from voltage measurement for AC Phase L1 from inverter before relay K200	ADC input	Samples the measurement output for voltage L1_U_INV w.r.t voltage M	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L2_U_INV	AN18	32	Analog input from voltage measurement for AC Phase L2 from inverter before relay K300	ADC input	Samples the measurement output for voltage L2_U_INV w.r.t voltage M	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L3_U_INV	AN11	23	Analog input from voltage	ADC input	Samples the measurement output for	All the time at a periodic rate (nominally 200 times	ADC IN

			measurement for AC Phase L3 from inverter Before relay K400		voltage L3_U_INV w.r.t voltage M	per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	Voltage
L1_U_MID	AN14	12	Analog input from voltage measurement for AC Phase L1 after relay K200	ADC input	Samples the measurement output for voltage L1_U_MID w.r.t voltage M	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L2_U_MID	AN15	21	Analog input from voltage measurement for AC Phase L2 after relay K300	ADC input	Samples the measurement output for voltage L2_U_MID w.r.t voltage M	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L3_U_MID	AN16	30	Analog input from voltage measurement for AC Phase L3 after relay K400	ADC input	Samples the measurement output for voltage L3_U_MID w.r.t voltage M	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L1_U_GRID	AN10	24	Analog input from voltage measurement for AC grid phase L1	ADC input	Samples the measurement output for voltage L1_U_GRID w.r.t voltage N_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L2_U_GRID	AN12	11	Analog input from voltage measurement for AC grid phase L2	ADC input	Samples the measurement output for voltage L2_U_GRID w.r.t voltage N_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L3_U_GRID	AN13	22	Analog input from voltage measurement for AC grid phase L3	ADC input	Samples the measurement output for voltage L3_U_GRID w.r.t voltage N_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN Voltage
L1_I_INV	AN0	13	Analog input from current measurement for AC inverter phase L1 before relay K200	ADC input	Samples the measurement output for current L1_I_INV	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT
L2_I_INV	AN1	14	Analog input from current measurement for AC inverter phase L2 before relay K300	ADC input	Samples the measurement output for current L2_I_INV	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT
L3_I_INV	AN2	15	Analog input from current measurement for AC inverter phase L3 before relay K400	ADC input	Samples the measurement output for current L3_I_INV	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT

L1_I_GRID	AN6	28	Analog input from current measurement for AC inverter phase L1 supplying to the grid (after relay K100)	ADC input	Samples the measurement output for current L1_I_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT
L2_I_GRID	AN8	27	Analog input from current measurement for AC inverter phase L2 supplying to the grid (after relay K101)	ADC input	Samples the measurement output for current L2_I_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT
L3_I_GRID	AN9	35	Analog input from current measurement for AC inverter phase L3 supplying to the grid (after relay K102)	ADC input	Samples the measurement output for current L3_I_GRID	All the time at a periodic rate (nominally 200 times per 50Hz or 60Hz cycle i.e. 10kHz or 12kHz).	ADC IN CURRENT
TEMP1	AN3	16	Analog input from NTC sensor attached to IGBT Heatsink	ADC input	Samples temperature measurement 1 from IGBT Heatsink If preset limit reached then fault condition alerted. IGBT Switching is shut down.	All the time at a periodic rate (nominally once every second). (might be subsequently configured as comparator interrupt TBD)	ADC IN TEMP
TEMP2	AN4	17	Analog input from NTC sensor attached to LCL Filter Heatsink	ADC input	Samples temperature measurement 2 From heatsink location close to main inductors on LCL filter. If preset limit reached then fault condition alerted. IGBT switching is shut down.	All the time at a periodic rate (nominally once every second). (might be subsequently configured as comparator interrupt TBD)	ADC IN TEMP
TEMP3	AN5	18	Analog input from NTC sensor measuring ambient temperature	ADC input	Samples temperature measurement 3 ambient temperature If preset limit reached then fault condition alerted. IGBT switching is shut down.	All the time at a periodic rate (nominally once every second). (might be subsequently configured as comparator interrupt TBD)	ADC IN TEMP
ITRIP_INV	RD6	54	Duty cycle output generating voltage input to current measurement ICs, setting over current trip level	GPIO (output only)	Sets voltage for over current level for the inverter. This will be the over current level for the local load.	All the time. Cycle frequency 10kHz Duty cycle varying from 10% to 90%	ITRIP SET
ITRIP_GRID	RC3	55	Duty cycle output generating voltage input	GPIO (output	Sets voltage for over current level supplied to the grid.	All the time. Cycle frequency 10kHz Duty cycle varying from 10% to 90%	ITRIP SET

			to current measurement ICs, setting over current trip level	only)			
L1_I_INV_OC/	RC4	50	Flag input for over current condition for L1_I_INV	GPIO (input)	Lo = over current condition for L1_I_INV	Interrupt highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L2_I_INV_OC/	RC5	51	Flag input for over current condition for L2_I_INV	GPIO (input)	Lo = over current condition for L2_I_INV	Interrupt highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L3_I_INV_OC/	RC7	36	Flag input for over current condition for L3_I_INV	GPIO (input)	Lo = over current condition for L3_I_INV	Interrupt highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L1_I_GRID_OC/	RD8	42	Flag input for over current condition for L1_I_GRID	GPIO (input)	Lo = over current condition for L1_I_GRID	Interrupt highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L2_I_GRID_OC/	RD9	40	Flag input for over current condition for L2_I_GRID	GPIO (input)	Lo = over current condition for L2_I_GRID	Interrupt - highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L3_I_GRID_OC/	RD14	39	Flag input for over current condition for L3_I_GRID	GPIO (input)	Lo = over current condition for L3_I_GRID	Interrupt - highest priority. Action: IGBT switching stops immediately. Relays opened	FLAG OC
L1_N_PWM	PWM1L	3	PWM to drive low IGBT on half-bridge for AC phase 1	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT
L1_P_PWM	PWM1H	2	PWM to drive high IGBT on half-bridge for AC phase 1	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT
L2_N_PWM	PWM2L	63	PWM to drive low IGBT on half-bridge for AC phase 2	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT
L2_P_PWM	PWM2H	62	PWM to drive high IGBT on half-bridge for AC phase 2	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT
L3_N_PWM	PWM3L	61	PWM to drive low IGBT on half-bridge for AC phase 3	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT
L3_P_PWM	PWM3H	60	PWM to drive high IGBT on half-bridge for AC	PWM output	Hi = IGBT ON	16kHz. Variable duty cycle	PWM IGBT

			phase 3				
250kHz_T2	PWM4L	64	PWM for IGBT drive power supply	PWM output	Hi = IGBT ON	250kHz. Variable duty cycle	PWM IGBT
250kHz_T1	PWM4H	1	PWM for IGBT drive power supply	PWM output	Hi = IGBT ON	250kHz. Variable duty cycle	PWM IGBT
U1_RXD	RC14	34	UART for RS485 (and initial debug)	UART RX (input)	RS-485 UART RX (also use for initial debug)		UART1
U1_TXD	RC15	31	UART for RS485 (and initial debug)	UART TX (input)	RS-485 UART TX (also use for initial debug)		UART1
U1_TXEN/	RB4	33	GPIO to enable RS485 for TX	RS485 TX enable	Lo = RS485 TX enable Hi = RS485 RX enable	Default = Hi	UART1
U2_RXD	RC0	4	UART2 RX	UART RX	UART RX for coms with DC-DC converter		UART2
U2_TXD	RC13	5	UART2 TX	UART TX	UART TX for coms with DC-DC converter		UART2
AUX1	RC8	37	LED502 drive/ spare GPIO	GPIO	Spare GPIO for indicator LED	Spare GPIO/ LED indication	Spare GPIO
AUX2	RD10	6	LED501 drive/ spare GPIO	GPIO	Spare GPIO for indicator LED	Spare GPIO/ LED indication	Spare GPIO
AUX3	RD11	45	LED500 drive/ spare GPIO	GPIO	Spare GPIO for indicator LED	Spare GPIO/ LED indication	Spare GPIO
AUX4	RD12	8	GPIO spare	GPIO			
AUX5	RB6	48	GPIO spare	GPIO			
AUX6	RB7	49	GPIO spare	GPIO			
PGED3	PGED3	43	Programming data				
PGEC3	PGEC3	44	Programming clock				
RESET/	MCLR/	7	Reset				

Table 2: Signals to the dsPIC

5 Top Level State Machine Description for Grid Inverter

A top level state machine for operation of the grid inverter is shown in figure 3. The following sections provide a description of intended operation.

5.1 Start and Idle States

Operation begins once the auxiliary power supplies are on and the control electronics for the grid inverter are powered.

The idle state is entered immediately after power-up, or a reset to the microcontroller has been asserted. The idle state is shown in the middle of the state machine diagram (figure 3).

In this state the microinverter is initiated and configured as required.

All relays are open. The IGBTs are not being switched and PWM outputs remain low.

RS-485 communication with the Melacs board is checked. UART communication with the DC-DC converter is checked.

Inputs from the Melacs board, necessary for operation, are received and acknowledged.

These inputs are listed below in table 3.

Symbol	Description	Type	Dynamic, Static, Command
AC_phases_out	Number of AC phases required	Restricted integer: 1,3 values allowed	Static
GRID	Supply to grid or not	Binary: 0 = no grid supply, 1 = supply to grid	Static
Ac voltage nominal	Nominal AC voltage (RMS)	Integer between 90Vac and 265Vac	Static
AC minimum voltage	AC voltage min (RMS)	Integer between 80Vac and 280Vac	Static
AC max voltage	AC Voltage max (RMS)	Integer between 80Vac and 280Vac	Static
Frequency	Nominal frequency (Hz)	Integer either 50Hz or 60Hz	Static
Min Frequency	Minimum allowed frequency	Floating point ranging from potentially 49Hz to 61Hz	Static
Max Frequency	Maximum allowed frequency	Floating point ranging from potentially 49Hz to 61Hz	Static
Anti-Islanding Operation	Which anti-islanding and grid fault behaviours are active.	Integers defining cases: Examples (TBD) 1 Default passive anti-islanding 2 Ground fault currents 3 UVRO 4 OVRO others	Static
Max power	Maximum allowable power to deliver to grid	Integer ranging from 0 to 10k	Static
CHARGING	Signal charging mode from the grid to main battery pack	Binary: 0 = not charging, 1 = charging.	Command
Test mode	Active = ready to supply power, test mode = system check before supplying power	Binary: 0 = test mode; 1 = active mode.	Command
Inverter On	OK to start inverter switching	Binary: 0 = test mode; 1 = active	Command
Power Set	Energy level set to deliver from inverter (when connected to grid)	Integer ranging from 0 to 10k	Dynamic (input from outer control loop)
Fault	System Fault notification from Melacs board	Binary flag	Dynamic (interrupt)

Table 3: Inputs in the idle state from Melacs Board

All the static inputs listed in table 3 are constants providing necessary information to set up correct operation and control when power is being delivered.

Three inputs are effectively commands: Charging, Test Mode, Inverter On. Only one of these inputs can be active at any one time. (If more than one is active for some reason then stay in idle state).

Two inputs are dynamic: Fault and Power Set.

Fault input active flags that a fault is present elsewhere in the system. This may be input at any time and in any state. Should this input be active then operation remains in the idle state. Should this input be active during any other state then operation moves to the idle state.

Power Set input determines the level of power the inverter should deliver when connected to the grid. It is an input from the outer control loop comprising the stirling engine, motor inverter and DC-DC converter. This input is updated on a periodic basis, nominally once every second (TBC). It will also appear as an input during the single-phase and 3-phase PLL current loop control states.

5.2 Local Load Connection

The states for local load connection are shaded in blue in figure 3. These states are entered when the necessary operational inputs listed in table 3 have been received, and the input indicating Grid is inactive and the input indicating Inverter On is active.

5.2.1 Check Status

A status check is undertaken initially. As such ADCs will need to be active.

This state checks for the following:-

1) That the relays are presently open.
2) That no faults are flagged.
3) That none of the temperature inputs are outside allowable limits.
4) That input DC voltage is at a level required for assigned operation.
5) That no grid is present.

If any of 1) through 4) of the above are false, then revert to the idle state.

If 5) is false then the detection of the grid needs to be communicated with the Melacs board. (If connection with the grid is subsequently required, then the Melacs board will instruct this and the inputs in table 3 will be updated). The state machine should stay in this current state until the type of connection is confirmed with the Melacs board.

The PWMs driving the local power supply for the IGBTs (PWM4L & H) are operated and a defined period of waiting occurs to allow the power supply to stabilise (nominally 1 second).

From this point the state machine splits, depending on whether the load is 3-phase or single phase.

In either case the inverter will use sinusoidal PWM modulation to generate the required single-phase or 3-phase output. At this point the relays are still open and so no power is being delivered.

Once the AC voltage waveform/s (U_INV) meet requirements for voltage, frequency and phase for a nominal number of cycles (nominally 50 cycles) then the inverter relays (K103, K200, K300 & K400) will be closed.

5.2.2 PLL Voltage Mode Control Loop

From this point there is a choice for operation for closed loop voltage control:-

1. Zero-Cross Detection Phase-Locked – Loop (ZCD-PLL) using Microchip Reference Design and Source Code [9]:
   Since the system is operating off-grid there is less possibility for noise and error on the supply. It is therefore possible to implement the ZCD-PLL control loop utilised by Microchip for their reference design, but using reference coefficients in place of grid ADC measurements. Microchip's PLL uses zero cross detect to control frequency and phase and adjusts PWM duty cycle in line with sinusoidal modulation coefficients.[4]

   It is understood that access to source code for the above is available from Microchip.

2. Synchronous Frame Phase-Locked-Loop (SF-PLL), being considered for grid connected PLL:
   Completing dq-zero transformation (using reference table data to simulate grid connection) may also be an approach. This method may be overly complex for local load connection. However, commonality with this proposed method of PLL for grid connection may yield advantages. More information relating to this PLL approach is covered in section 5.4.

At the time of writing it is proposed that method 1 above is pursued, at least initially. This should be quickest to implement and will support project development and hardware verification.

Method 2 may have advantages, however, providing as much commonality as possible between control loops used for local load and grid connection.

5.2.3 Faults and Parameters Outside Limits

If at any time a fault condition is raised or the AC voltage falls outside the limits for voltage, frequency or phase (for a defined period – nominally 10 cycles) then operation moves directly to the idle state.

5.2.4 Grid Appearing

If the grid should appear at any time during these states, then this information should be communicated to the Melacs board.

[4] Details provided in Microchip application note AN1444

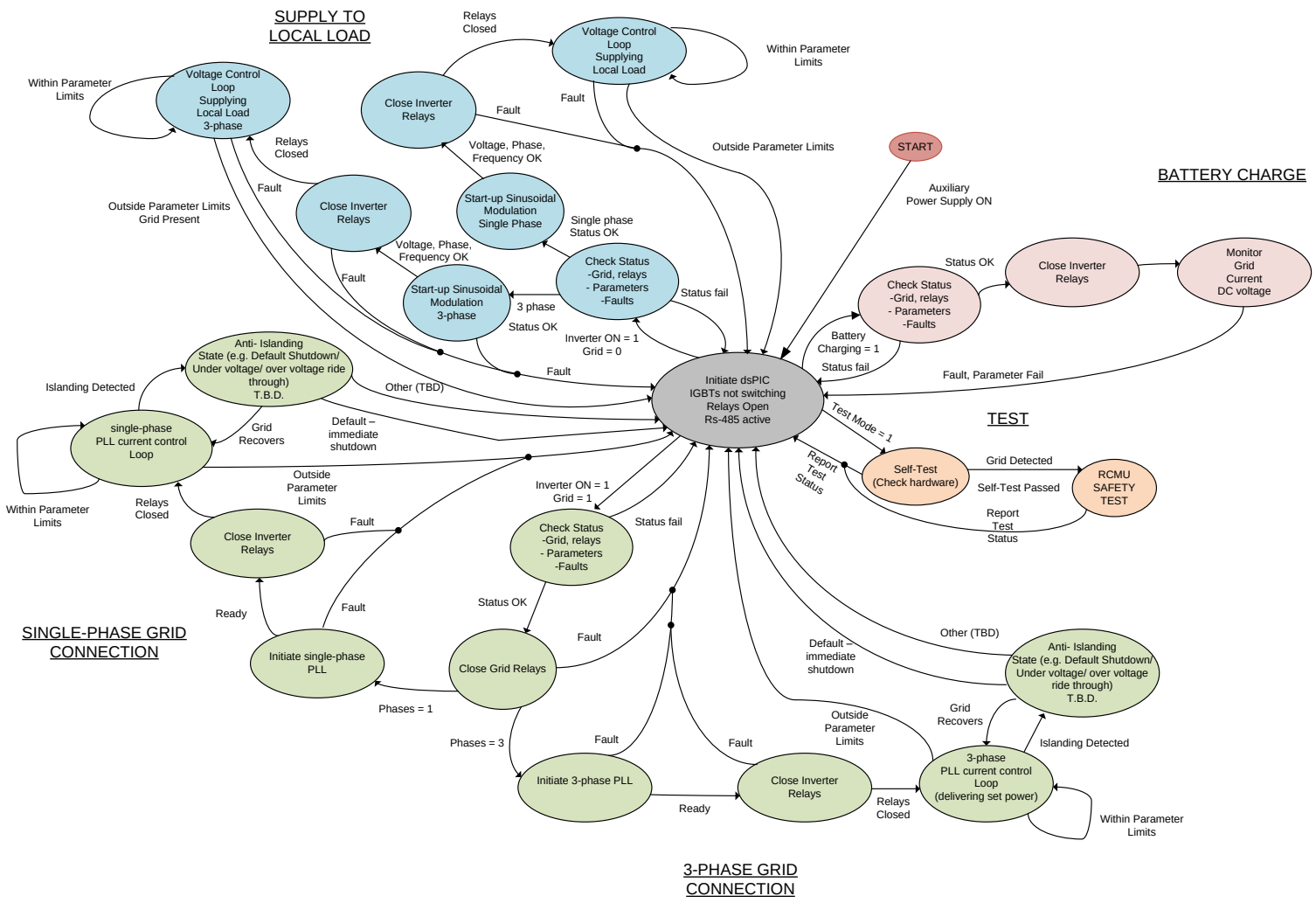


Figure 3: Top Level State Machine for Grid Inverter

5.3 Grid Connection

The states for single phase and 3-phase grid connection are shaded green in figure 3. These states are entered when the necessary operational inputs listed in table 3 have been received, and the grid input indicating Grid and the input indicating Inverter On are both active.

5.3.1 Check Status

A status check is undertaken initially. As such ADCs will need to be active.

This state checks for the following:-

1) That the relays are presently open.
2) That no faults are flagged.
3) That none of the temperature inputs are outside allowable limits.
4) That input DC voltage is at a level required for assigned operation.
5) That the grid and expected no. of phases are present

If any of 1) through 5) of the above are false, then revert to the idle state and report to the Melacs board.

The PWMs driving the local power supply for the IGBTs (PWM4L & H) are operated and a defined period of waiting occurs to allow the power supply to stabilise (nominally 1 second).

The grid connection relays are then closed (K104, K100, K101, K102) and power is supplied to the local load from the grid.

From this point the state machine splits, depending on whether the output required is 3-phase or single phase.

5.4 Phase Locked Loop (PLL) for 3-Phase Grid Connection

When connected to grid, PLL control is required to synchronise with the grid. Output voltage from the inverter is matched slightly higher with grid voltage (since the grid will essentially dominate) and output current is controlled. The output current must be sinusoidal, with harmonic content low and synchronised with the grid in frequency and phase. In this sense control differs from the local load connection. Under local load connection output voltage is controlled under varying load demand. Output current follows load demand under local load connection.

In considering an appropriate PLL routine for 3-phase grid connection the reader is referred to [2]. A control solution which has efficacy, but also design simplicity is sought. Under these considerations it is recommended that Synchronous Frame Phase Locked Loop (SF-PLL) is used.

SF-PLL may also be referred to as dq-transformation PLL. The method provides good performance for distortion insensitivity and adaptive frequency and range. It performs less well than other, more sophisticated, control schemes when the 3-phase grid is out of balance.

A general diagram of the control structure for SF-PLL is shown below in figure 4.

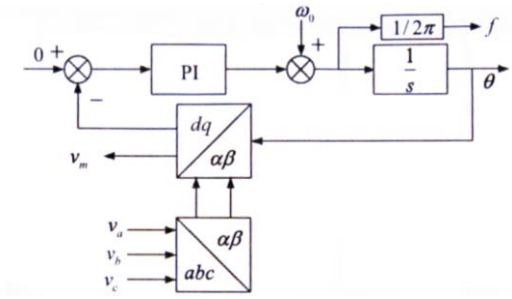


Figure 4: SF-PLL [2]

In this example for synchronising voltage components, vector components for each voltage phase are sampled using the ADCs (Va, Vb, Vc).

Initially, these components are converted to a stationary reference frame using the Clarke Transform [6], [7]. This produces a stationary reference frame with orthogonal components (α & β) varying at frequency ω. The next step is to complete the Park transform which converts the stationary reference plane to a rotating reference plane with two DC components Vd and Vq.

When locked to phase and frequency, and when the 3 phases Va, Vb and Vc are in balance, Vd = Vm, i.e. the peak AC voltage, and Vq = 0. When phase is out of lock Vq has a value which is proportional to the phase error. This property can therefore be used to form a PLL and lock onto and synchronise SPWM modulation with the 3-phase grid voltages.

A visual example of the above is shown below in figure 5.

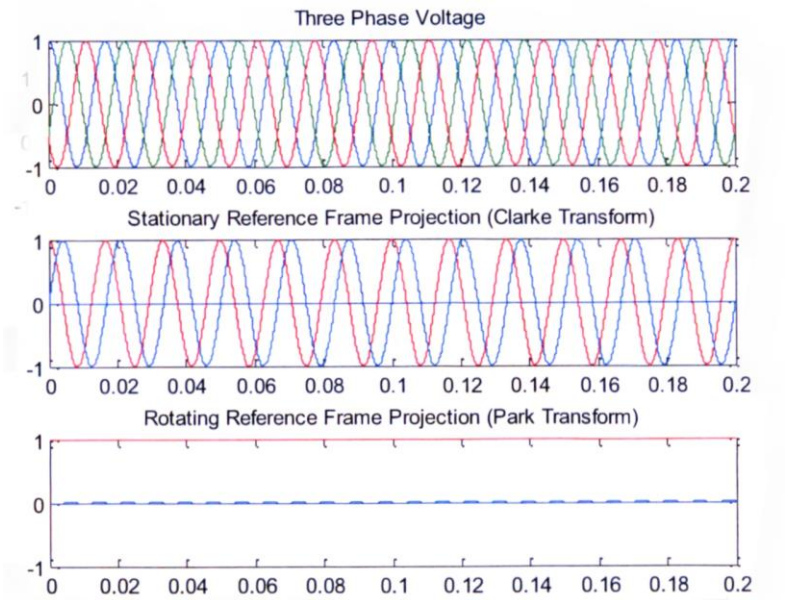


Figure 5: Simulation outputs showing
a) 3-phase Voltage waveform
b) orthogonal outputs from the Clarke Transform and
c) Vd (red) and Vq (blue) outputs for the Park Transform [6].

Figure 6 provides a further visual example of a PLL using the dq components to track a phase jump of 90°.

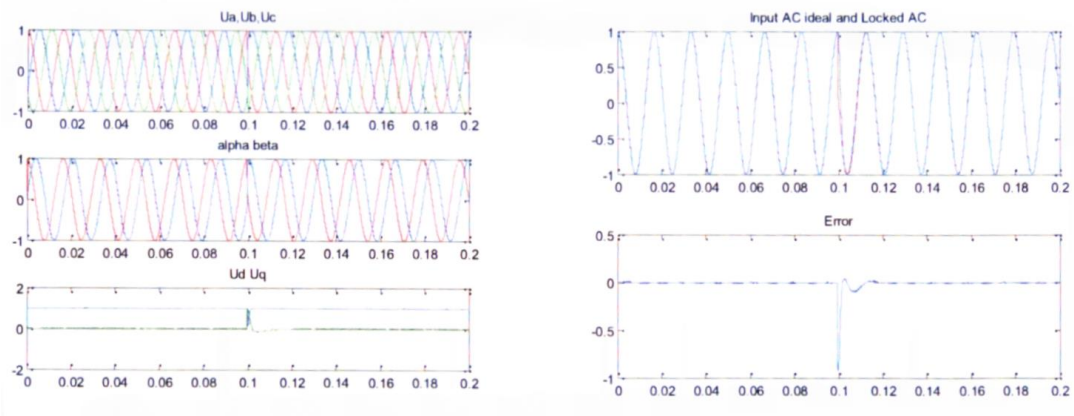


Figure 6: Example of simulated waveforms of a PLL using dq transform reacting to a 90° phase jump. The waveforms on the right shows the input AC and locked AC response and the phase error [6].

The Park and Clarke transforms can be reversed. In this way the Vq and Vd components can be transformed back to provide sinusoidal reference voltages for Sinusoidal PWM modulation of the half-bridges. This property is explained in [5] &[7].

In terms of a classic linear description of a PLL, which includes a phase detector, filter and VCO, shown below in figure 7, and comparing with figure 5, then the dq transform stage performs the function of the phase detector, the PI stage performs the function of the loop filter and the multiplier and integrator combined performs the function of the VCO.

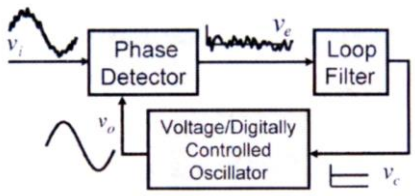


Figure 7: Linear PLL [1]

References which comment in detail on designing SF-PLL [1],[4],[5],[6] stress the need to model PI and integrator stages carefully and chose Q format word length carefully for optimisation. It may therefore be necessary to model functionality using tools such as SIMULINK in order to define an algorithm sufficiently accurately for intended operation.

In the state machine diagram, the state defined as ‘initiate 3-phase PLL’ relates to setting up the SF-PLL phase lock loop and delivering output voltage synchronised with the grid.

When connected to the grid, phase currents will also need to be synchronised and in phase with voltage so that unity power factor can be achieved. This requires further dq transform operation and a more complex overall control scheme in addition with SF-PLL for voltage. Only a few papers have been found to date which cover this area of design [3], [5]. Figure 8 below is a block diagram for a generalised control loop.

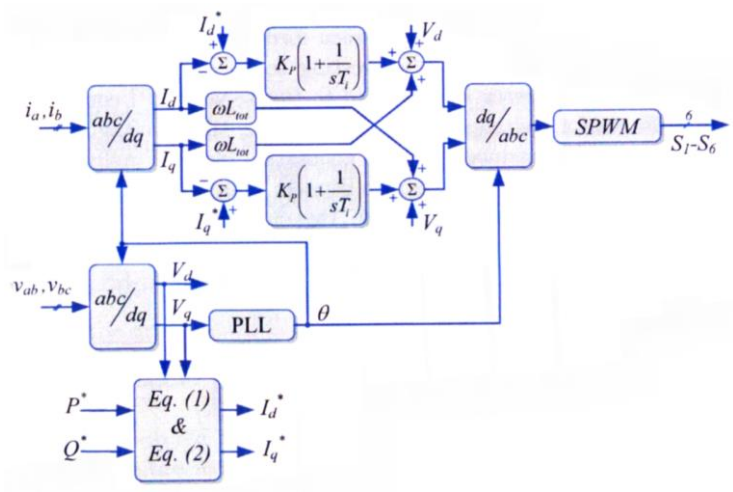


Figure 8: Generalised view of grid connected current and voltage control [5].

Within this generalised control scheme, the voltage SF-PLL is indicated, alongside additional blocks to control output current.

3-phase currents would be monitored using the ADCs in the dspic and these outputs input to a separate dq transform operation. If unity power is delivered, i.e. current synchronised and in phase with voltage, then the Id component is equal to the peak current. The Iq component will be zero. In terms of our intended operation, then unity output power is desirable. As such the Iq* component in figure 8 above would be zero. The Id* component would be set to the desired level to deliver output power (this would be a value derived from the Power Set input from the Melacs control board).

The full details around this control scheme require more study and understanding at the time of writing. As such I do not appreciate how some of the components within figure 8 contribute to overall control. For example, the relevance of the multiplier functions feeding into the inverse dq transform are not, as yet, understood.

The primary intention in this section is to indicate direction and provide some clarity on work that needs to be completed to reach intended operation. Furthermore, it is believed that development of the current control PLL can be conducted in parallel with the overall state machine development, since the software needed to progress and verify the hardware design does not necessarily require PLL functionality.

In the state machine the overall current control scheme indicated above would be implemented once the voltage PLL was stable and after the inverter relays were closed.

If operation of this control loop delivers current and voltages which are outside parametric limits, then operation would move to the default idle state. If islanding is detected then operation would move to the selected anti-islanding operation. More on this is discussed in section 5.6.

5.5 Single Phase PLL for Grid Connection

According to literature [2],[8] SF-PLL may be applied to single-phase grid connection by -passing the need for the Clarke transform and generating artificial orthogonal components for the Park Transform. Quoting [8]:-

> ‘The detection method used in this implementation for a single-phase inverter is based on a synchronous reference frame PLL. While in three-phase inverters the use of DQ based PLL is quite common, for single-phase inverters, the necessity of a virtual bi-phase system arises. In fact, to create a rotating DQ reference, starting from a stationary frame, at least two independent phases are required. This problem is overcome with the creation of a virtual voltage, V ⇓ , phase-

shifted with respect to the real grid voltage, Va , of 90°. This task may be easily accomplished with firmware.'

As such it is envisaged that algorithms developed for 3-phase operation may be readily modified for single-phase operation.

The states defined for the single-phase grid connection therefore follow in common sequence to those discussed in section 5.4.

5.6 Anti-Islanding

Islanding is the continued operation of the inverter when the grid has been removed intentionally, by accident or by damage.

When connected to the grid and there is a fault on the grid whereby supply is interrupted, power will potentially still be supplied to the grid from local generators (e.g. this system). In many cases this is undesirable for grid operators, for system stability and safety reasons. Under grid fault conditions then supply must be disconnected from the grid and switched to the local load. This behaviour is referred to as anti-islanding.

The simplest operation requires that power feed to the grid is disconnected once grid voltage or frequency drops outside allowable limits. This type of operation is preferred by Inresol. Generally, this type of operation is termed 'passive anti-islanding'.

Future Electronics is also aware that in some regions requirements for anti-islanding are being reviewed, potentially requiring different behaviour should grid voltage and frequency values measured indicate islanding [10]. These include:-

- Voltage Ride Through: is the capability of electric generators to stay connected for short periods of lower or higher electric network voltage. It is needed at distribution level to avoid that a short circuit or transient event on HV or EHV level will lead to a widespread loss of generation.
- Frequency Ride Through: similar to above, due to transient variations in frequency.

In the state machine diagram provision has been made for a state to cover anti-islanding operation, should needs subsequently change.

The default requirement currently is for passive anti-islanding operation, whereby, when parameters fall outside operating limits operation moves immediately to the idle state.

5.7 Battery Charge

As identified in section 3.3, there is a need during cold start to allow the main battery to be charged from an external AC connection.

The states shaded pink in figure 3 cover battery charge operation. These states are entered when the necessary operational inputs listed in table 3 have been received, and the input indicating Grid is inactive and the input indicating Charging is active.

5.7.1 Check Status

A status check is undertaken initially. As such the ADCs will need to be active. The state checks the following:-

1. That the relays are open.
2. That a grid connection is present on any of the AC phase connections.
3. That grid voltage is large enough to support battery charge.
4. That no faults are flagged.
5. PWM outputs are low and IGBTs are not switching.

If any of the above are false, then operation returns to the idle state.

Output from the check status state is communicated to:-

1. The DC-DC converter via UART.
2. The Melacs board.

The grid inverters are then closed. There is then a pause for several seconds (nominally 5 seconds) before the inverter relays are closed.

Grid voltage and currents, inverter voltage and currents and temperatures are monitored continuously and reported directly to the DC-DC converter via UART.

If monitored parameters fall outside allowable limits then relays are opened and operation returns to the idle state.

5.8 Self-Test

The states shaded sand colour in figure 3 cover self-test operation. These states are entered when the necessary operational inputs listed in table 3 have been received, and the input indicating test mode active.

Operation of a self-test return has yet to be discussed in detail. Details for this operation are tbd at the time of writing.

6 References

1. Freijedo, F.D. et al. 'An Optimised Implementation of Phase Locked Loops For Grid Applications', IEEE Transaction On Instrumentation and Measurement, Vo. 60, No. 9 Sept. 2011.
2. Guo Xiao-Qiang et al. (Yanshan University) 'Phase Locked Loop and Synchronisation Methods for Grid-interfaced Converters: a review', Przeglad Elektrotechniczny (Electrical Review) ISSN 0033-2097, Apr 2011.
3. Blaabjerg, F. et al. 'Overview of Control and Grid Synchronisation for Distributed Power Generation Systems', IEEE Transactions on Industrial Electronics, Vol. 53, No. 5, Oct 2006.
4. Ogren, Jim. (Uppsala Universitet) 'PLL Design For Inverter Grid Connection: Simulations for ideal and non-ideal grid conditions', July 2011.
5. Sarajian, Sajad. 'Design and Control of grid Interfaced Voltage Source inverter with output LCL Filter', Department of Electrical Engineering, Islamic Azad University, Iran, June 2014.
6. Bhardwaj, Manish. 'Software Phase Locked Loop design Using C2000 Microcontrollers for Three Phase Grid Connected Applications', Texas Instruments, Application Report, November 2013.
7. https://en.wikipedia.org/wiki/Direct-quadrature-zero_transformation
8. Application Note AN3095, '3kW Grid-Connected PV System', ST Microelectronics, Nov 2012.
9. Alex Dumais & Sabarish Kalyanaraman, 'Grid-connected Solar Microinverter Reference Design', Application Note AN14444, Microchip Inc. 2012.
10. Walling, R. & Ellis, A., 'Implementation of Voltage and Frequency Ride-Through Requirements in Distributed Energy Resources Interconnection Standards', Sandia National Laboratories, New Mexico, April 2014.